

Econ 646 Spring 2011
Second Exam (April 25th)

You will derive Ball's (1994) result that with a sticky price New Keynesian Phillips curve, disinflation causes booms. The model is continuous time, summerized by the following three equations:

$$p_t = \frac{1}{T} \int_0^T p_{t-s}^* ds \quad (1)$$

$$p_t^* = \frac{1}{T} \int_0^T E_t[m_{t+s}] ds \quad (2)$$

$$y_t = m_t - p_t \quad (3)$$

Variables are expressed in log. They are interpreted as follows. Equation (1) describes how aggregate price p_t is determined, where p_{t-s}^* is the price set by individual firms which set their prices at time $t-s$. The underlying assumption is that each firm resets its prices with every T interval, and in the aggregate, equal share of firms are resetting their prices at every point in time. Notice that these assumptions are analogous to the ones we saw in homework 7. Equation (2) describes individual firm's pricing decision, where m_{t+s} is nominal money stock in period $t+s$. This pricing decision is optimal in a certain environment. Money supply is controlled by the central bank. Finally, we assume that the quantity equation (3) holds, where y_t is output. It summerizes demand side of the economy.

Time index t can take negative number. In particular, $t=0$ does not mean the beginning of the history, but it is just a point in time. Answer the following questions.

1. Suppose that initially $m_t = gt$ ($g > 0$), and that $E_t[m_{t+s}] = (t+s)g$.

- (a) Solve for p_t^* .

(HINT: remember $\int_a^b s ds = \left[\frac{s^2}{2} \right]_a^b = \frac{b^2 - a^2}{2}$.)

- (b) Show that $p_t = gt$ and therefore $y_t = 0$.

- (c) In a diagram, show the path of p_t^* , p_t , and y_t , with time t on the horizontal axis. Clearly mark what values these variables take at $t=0$.

2. Suppose at time $t=0$, the central bank announces that it is steadily reducing money growth to 0 over the next T of time. Thus $m_t = t[1 - (t/2T)]g$ for $0 < t < T$, and $m_t = gT/2$ for $t \geq T$. The change is unexpected, so that prices set before $t=0$ (that is, p_t^* for $t < 0$) are as in part (1).

- (a) In a diagram, show the new annouced path of m_t , with time t on the horizontal axis.

- (b) Show that if $p_t^* = gT/2$ for all $t > 0$, then $p_t = m_t$ for all $t > 0$, and thus that output is the same as it would be without the change in policy.

(HINT: you want to calculate

$$p_t = \frac{1}{T} \int_0^T p_{t-s}^* ds = \frac{1}{T} \left[\int_0^t p_{t-s}^* ds + \int_t^T p_{t-s}^* ds \right] \text{ for } 0 < t < T$$

and

$$p_t = \frac{1}{T} \int_0^T p_{t-s}^* ds \text{ for } t \geq T$$

Note that p_t^* for $t < 0$ is set before the announcement, and therefore, they are set according to what you found in question (a) of part 1. This question asks what is p_t if we assume that prices set after the announcement are $p_t^* = gT/2$ for all $t > 0$.)

- (c) Are the prices actually set by firms for $0 < t < T$ (i.e., $p_t^* = \frac{1}{T} \int_0^T E_t[m_{t+s}] ds$ for $0 < t < T$) more than, less than, or equal to $gT/2$?
(HINT: be careful that $E_t[m_{t+s}]$ for $t > 0$ is NOT equal to $(t+s)g$ anymore. They are updated after the announcement.)
- (d) How does output during the period $(0, T)$ compare with what it would be without the change in policy?
- (e) Give some intuition for your answer in the previous question.
- (f) Are the prices actually set by firms for $T \leq t \leq 2T$ (i.e., $p_t^* = \frac{1}{T} \int_0^T E_t[m_{t+s}] ds$ for $T \leq t \leq 2T$) more than, less than, or equal to $gT/2$?
- (g) How does output during the period $(T, 2T)$ compare with what it would be without the change in policy?
- (h) Give some intuition for your answer in the previous question.